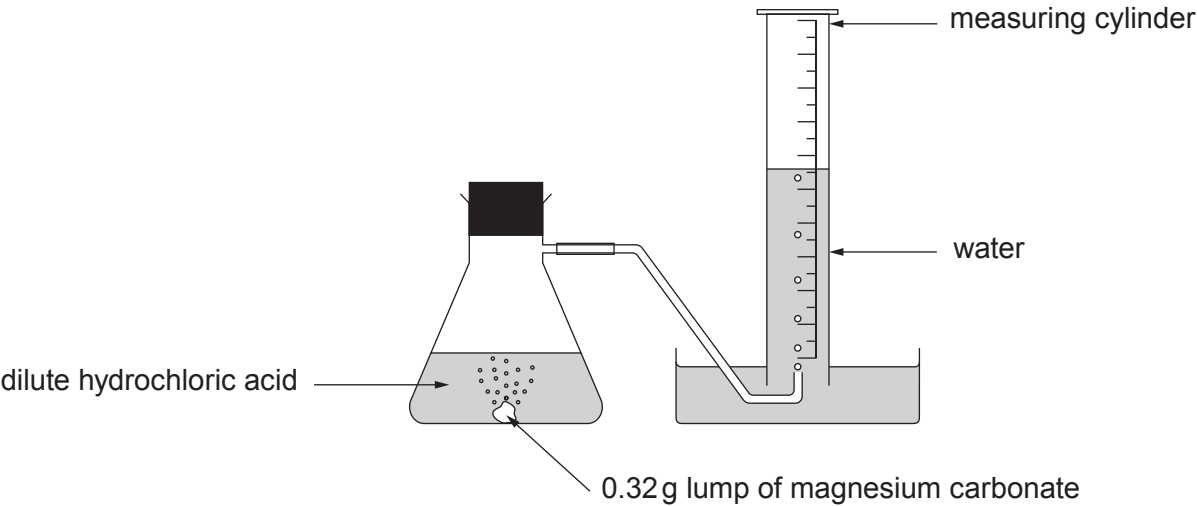


9. A student carried out an experiment to investigate the speed of the reaction between a **lump** of magnesium carbonate of mass 0.32 g and excess dilute hydrochloric acid at 20 °C.

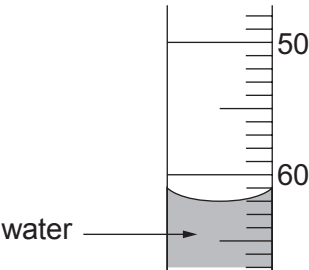
Magnesium carbonate reacts with dilute hydrochloric acid forming carbon dioxide gas. The total volume of carbon dioxide formed was recorded every 5 minutes for 40 minutes.



The results are shown below. The result for 15 minutes is missing.

Time (minutes)	0	5	10	15	20	25	30	35	40
Volume of carbon dioxide formed (cm ³)	0	20	41		79	83	90	90	90

- (a) Use the diagram below to find the volume of carbon dioxide gas formed after 15 minutes. [1]

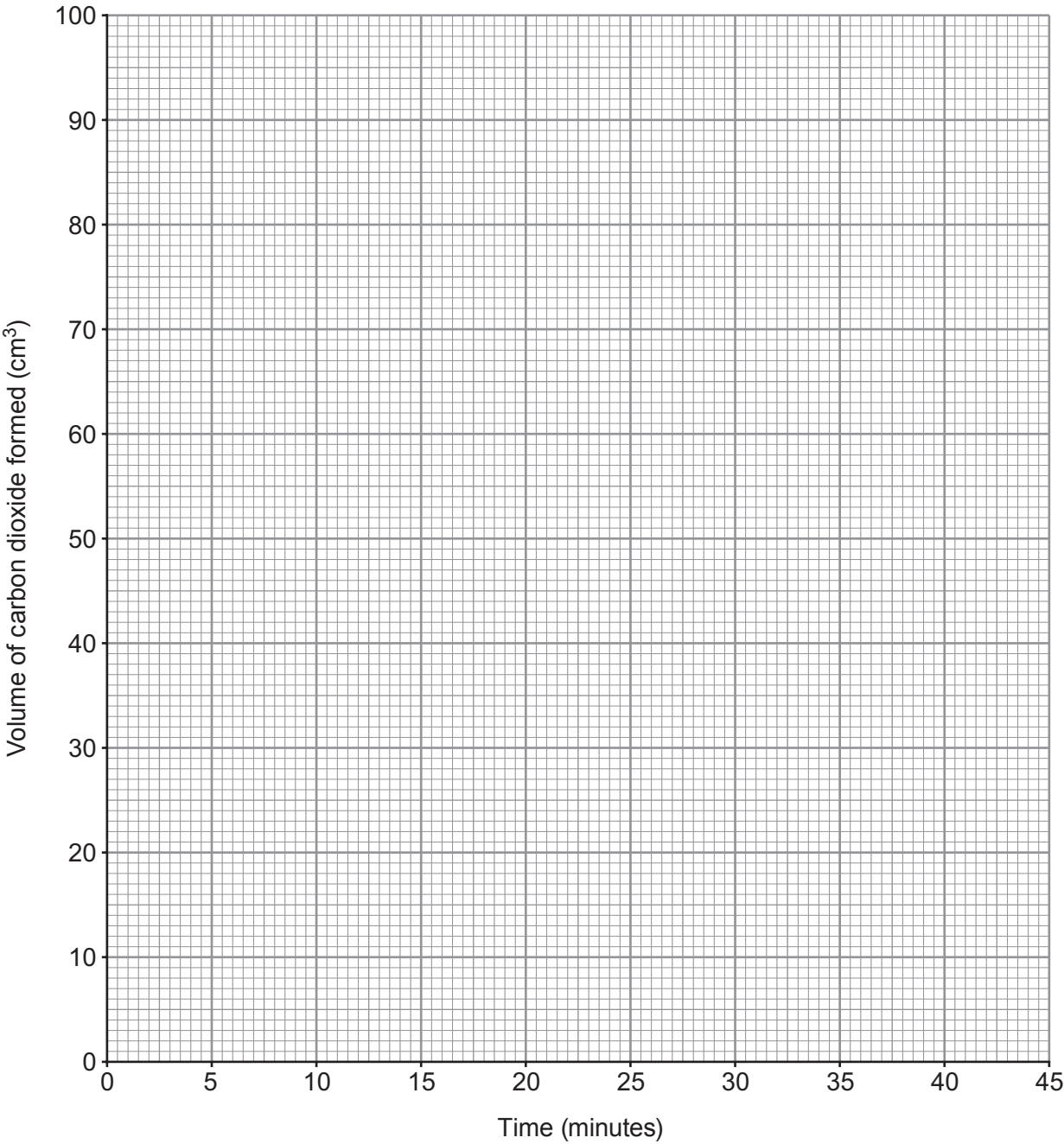


volume of carbon dioxide = cm³

Examiner
only

- (b) Plot the results from the table, including your answer to part (a), on the grid below. Draw a suitable line and **label this graph X**.

[3]



- (c) Sketch the graph you would expect if the experiment were repeated using 0.32 g of magnesium carbonate **powder** instead of the lump of magnesium carbonate. **Label this graph Y**.

[2]

(d) State and explain, using particle theory, the effect of **increasing** the concentration of the hydrochloric acid. [3]

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only

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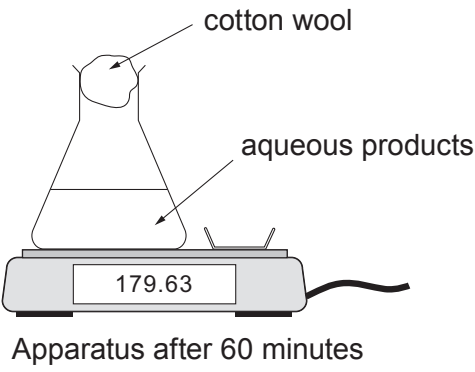
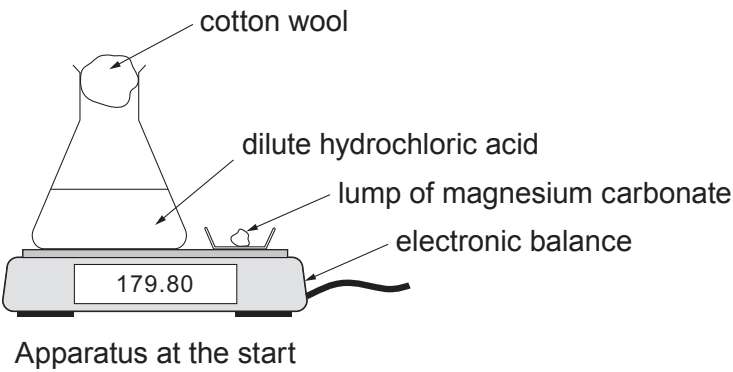
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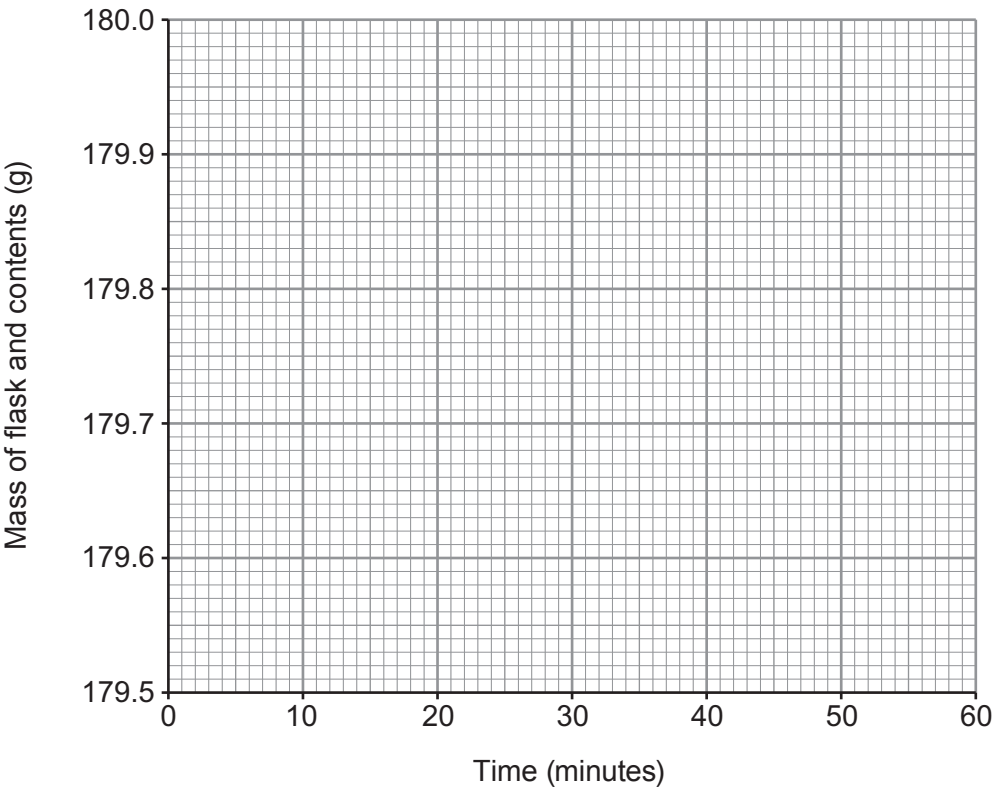
.....

- (e) The student investigated the same reaction using a different apparatus. A lump of magnesium carbonate was added to excess dilute hydrochloric acid at 20 °C.



The change in mass was recorded for 60 minutes and displayed as a graph on a computer screen. The reaction took 40 minutes to complete.

Sketch the graph you would expect to see. [2]



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		

THE PERIODIC TABLE

Group

2



4



6

2

0

1 H Hydrogen 1																			
7 Li Lithium 3	9 Be Beryllium 4									11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10				
23 Na Sodium 11	24 Mg Magnesium 12									27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18				
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36		
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54		
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86		
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89																	
Key																			

Key

relative atomic mass

Ar	Symbol	Name	Z
----	--------	------	---

Surname	Centre Number	Candidate Number
Other Names		0

GCSE – **NEW**

3410U10-1



S18-3410U10-1

CHEMISTRY – Unit 1:
Chemical Substances, Reactions and
Essential Resources

FOUNDATION TIER

WEDNESDAY, 13 JUNE 2018 – MORNING

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **8** is a quality of extended response (QER) question where your writing skills will be assessed.

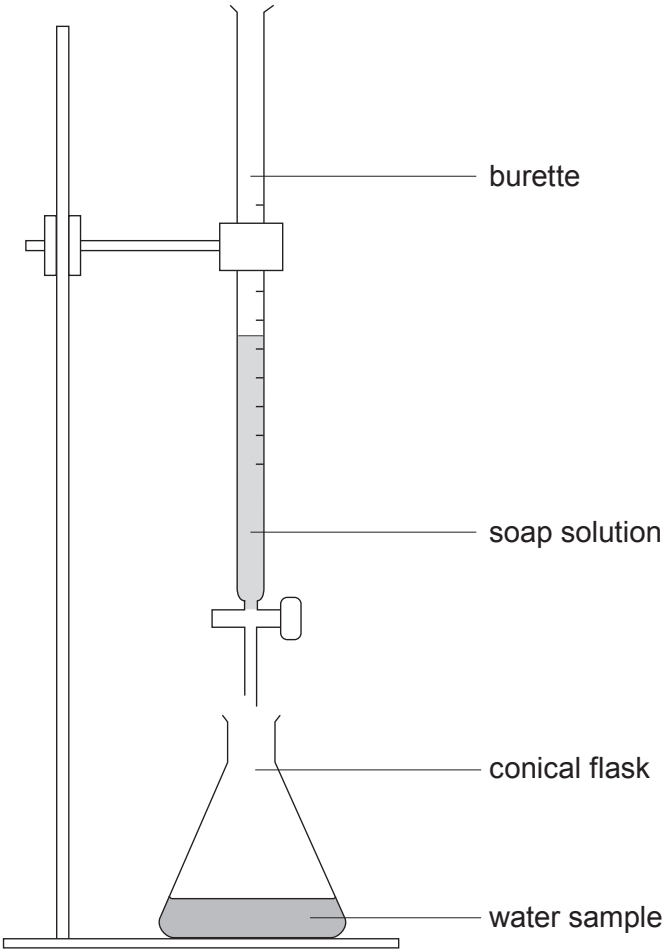
The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	6	
3.	7	
4.	8	
5.	10	
6.	9	
7.	5	
8.	6	
9.	6	
10.	9	
11.	5	
Total	80	

3410U101
01

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9. Water samples **A**, **B**, **C** and **D** were tested for hardness using the apparatus shown.



Soap solution was added 1 cm³ at a time to each sample and the volume required to produce a permanent lather on shaking was recorded. Each sample was tested before and after boiling. The results are shown in the table.

Water sample	Volume of soap solution required (cm ³)	
	Before boiling	After boiling
A	1	1
B	10	10
C	15	1
D	15	8



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(a) (i) State which water sample contains **only** temporary hardness. Explain your answer. [2]

Water sample

Explanation

(ii) Give **one** similarity in the composition of temporary and permanent hard water. [1]

(b) Discuss the benefits and drawbacks of living in a hard water area. [3]

6



9. (a) The following table shows some information about Group 1 elements.

Metal	Melting point (°C)	Boiling point (°C)	Density (g/cm ³)	Reaction with chlorine
lithium	180	1342	0.54	reacts slowly to make a white salt
sodium	97	883	0.97	burns vigorously with a yellow flame to make a white salt
potassium	63	759	0.88	reacts violently to make a white salt
rubidium	39	688	1.53	explosive reaction
caesium	28	671	1.93	explosive reaction

(i) Describe the trend in density going down the group. [1]

.....

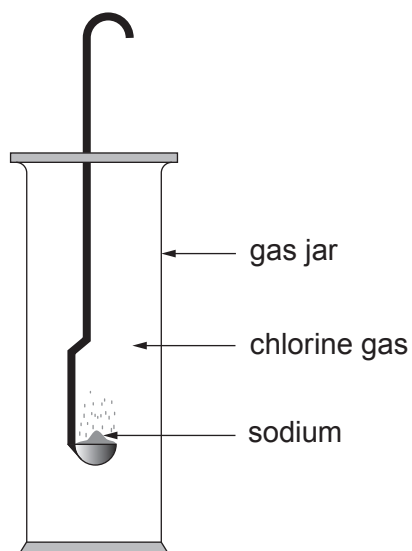
(ii) Explain the difference in reactivity down the group in terms of electronic structure. [2]

.....

.....



- (b) The apparatus below can be used to demonstrate the reaction between sodium and chlorine.



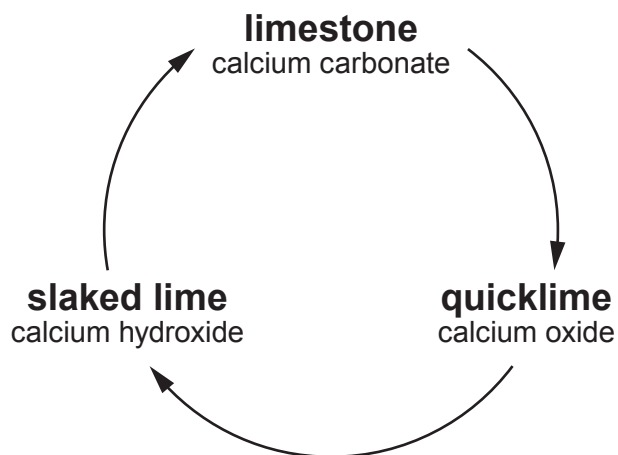
- (i) Apart from the use of safety goggles, state **one** safety precaution that needs to be followed when using **each** of these elements. [2]

Element	Safety precaution
sodium	
chlorine	

- (ii) Complete and balance the symbol equation for the reaction that takes place between sodium and chlorine. [2]



9. Limestone is a rock which consists mostly of calcium carbonate. The diagram shows a cycle of reactions involving limestone.



- (a) (i) When limestone is heated, calcium carbonate is converted to calcium oxide and carbon dioxide.

I. State the name for this type of reaction. [1]

.....

II. Write a balanced symbol equation for the reaction. [2]

..... → +

(ii) State what must be added to calcium oxide to form calcium hydroxide. [1]

.....

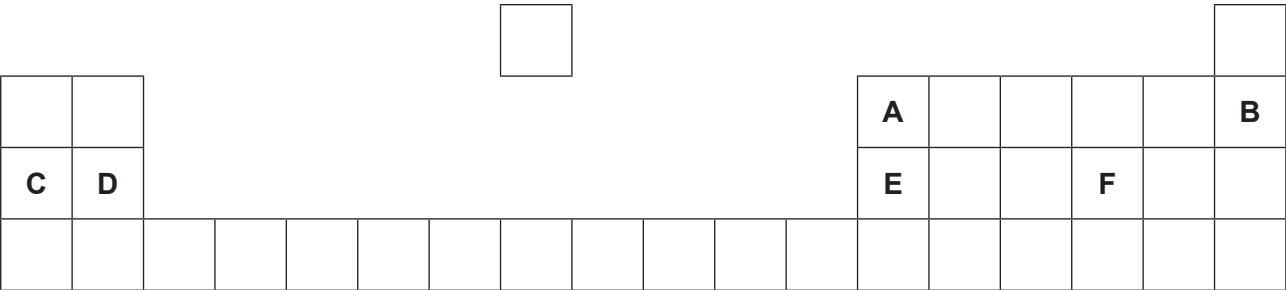
(b) Give **one** use of limestone in the construction industry. [1]

.....



9. (a) The following diagram shows an outline of part of the Periodic Table.

The letters shown are NOT the chemical symbols of the elements.



Choose **letters** from the diagram to complete the table below.

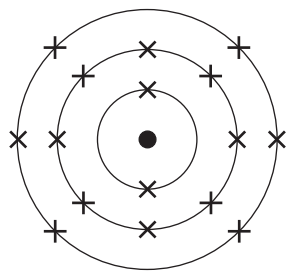
[4]

	Letter
The element in Group 3 and Period 2	
The element which has 10 protons in its nucleus	
The element with the electronic structure 2,8,6	
The element which forms a 2+ ion	



Examiner
only

(b) The diagram below shows the electronic structure of an element in the Periodic Table.



In the space below, draw a diagram to show the electronic structure of the element which lies directly **above** it.

[1]

(c) The table shows information about atoms **X**, **Y** and **Z**.

Atom	Symbol	Number of protons	Number of neutrons	Number of electrons
X	³¹ X ₁₅		16	15
Y	³⁹ Y ₁₉	19		19
Z	⁴⁰ Z ₁₉	19	21	

(i) Complete the table.

[3]

(ii) Underline the term used to describe atoms **Y** and **Z**.

[1]

ions inert insoluble isotopes

9

